INTERNATIONAL STATISTICAL ECOLOGY CONFERENCE 2027

MODELING SOWING TIMING UNDER CLIMATE VARIABILITY USING RANDOM WALKS ON GRAPHS

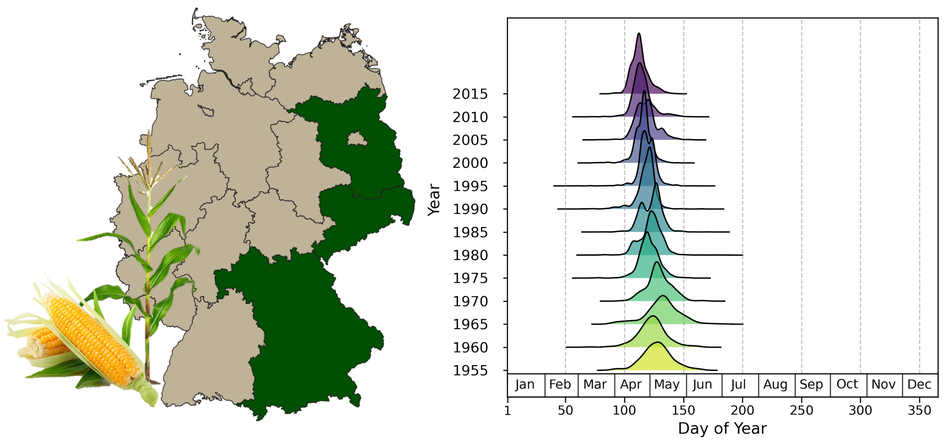
Isaac Vázquez Mendoza[1], Magali Arellano Vázquez[1].

[1]INFOTEC Centro de Investigación e Innovación en Tecnologías de la Información y Comunicación.

Introduction

The progression and duration of crop developmental stages are largely driven by environmental conditions [9]. Climate change is expected to alter the timing and duration of these stages [17, 16]. According to Baum *et al.* (2019), sowing dates may account for up to 70% of maize yield variability. Depending on geographical location, maize sowing dates may remain unchanged or shift earlier or later in the year [13]. Climate change projections suggest that northern European regions, including Germany, may become increasingly suitable for maize cultivation [14].

In Germany, maize sowing dates have shifted towards earlier dates, as illustrated in Figure 1b. This shift has been associated with climate change and has intensified since the late 1980s [5]. In this context, we developed a probabilistic model to predict maize sowing windows for each of the three German federal states considered in this study: Bavaria, Brandenburg, and Saxony, see Figure 1a- A region encompassing approximately 55% of the country's area devoted to major crops [6].



(a) Area of study. From top to bottom: Brandenburg, Saxony, and Bavaria.

(b) Distributions of maize sowing dates in Germany.

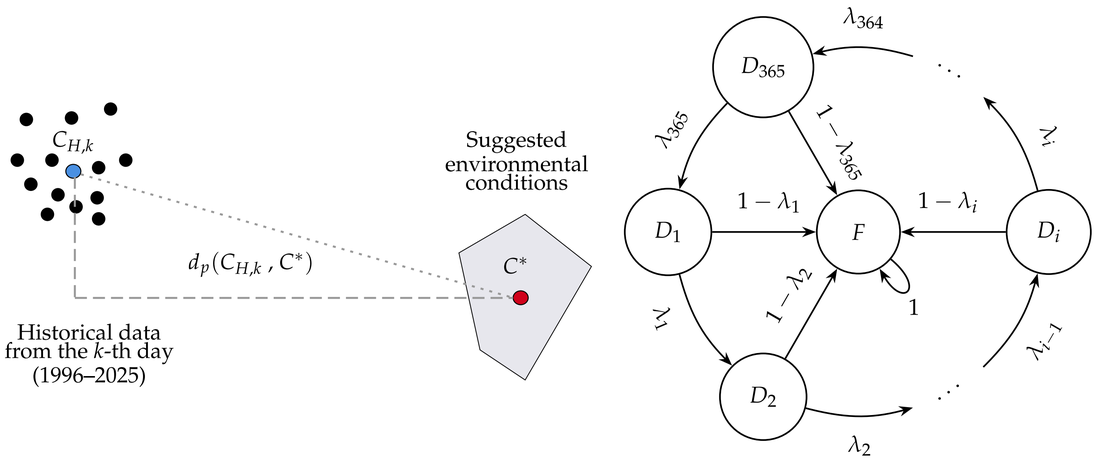
Figure 1: Statistical comparisons.

Methodology

Given $p \in \{1,2\}$, let $d(\cdot,\cdot) := d_p(\cdot,\cdot)$. Therefore, for each $k \in \{1,2,\ldots,365\}$

$$\lambda_k = 1 - \frac{d(C_{H,k}, C^*)}{\max\limits_{l\in\{1,\ldots,365\}} \{d(C_{H,l}, C^*)\}}. \tag{1}$$

denotes the transition probability.



(a) p-value of Kolmogorov-Smirnov test at different population sizes.

(b) χ^2 goodness-of-fit for Luria-Delbruck distribution.

Figure 2: Statistical comparisons.

Agroclimatic variable	Interval used in this study
Mean air temperature	11.10 ± 2.51 °C
Mean air relative humidity	65–85 %
Mean soil temperature at a depth of 5 cm	8–10 °C
Mean soil moisture in the 0–10 cm soil layer	80–90 %

Table 1: Daily

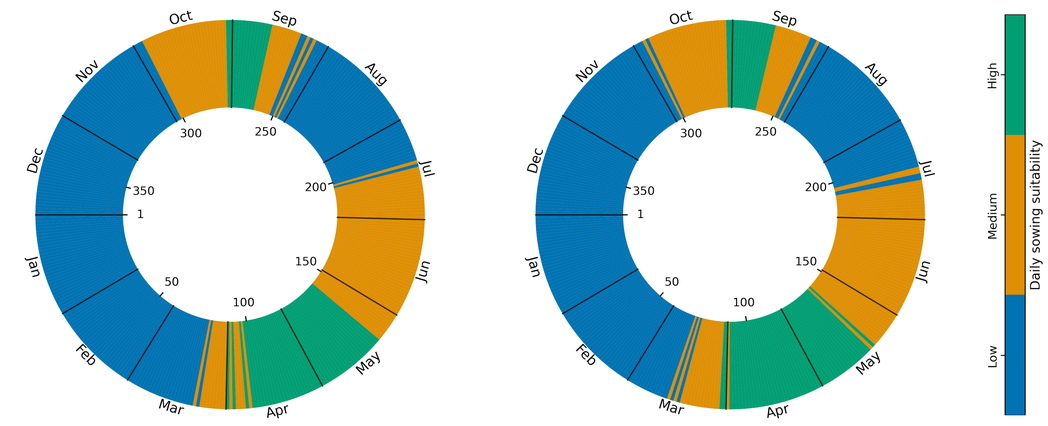
Results



(a) p-value of Kolmogorov-Smirnov test at different population sizes.

(b) χ^2 goodness-of-fit for Luria-Delbruck distribution.

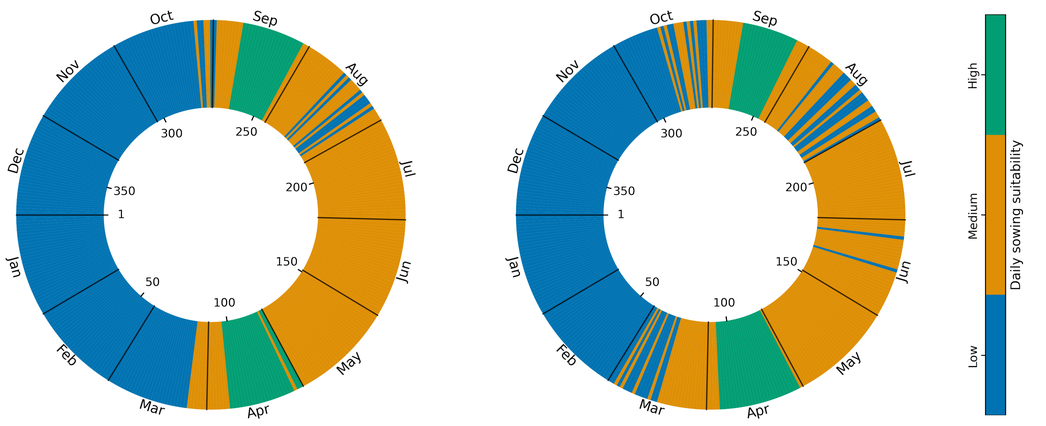
Figure 3: Statistical comparisons.



(a) p-value of Kolmogorov-Smirnov test at different population sizes.

(b) χ^2 goodness-of-fit for Luria-Delbruck distribution.

Figure 4: Statistical comparisons.



(a) p-value of Kolmogorov-Smirnov test at different population sizes.

(b) χ^2 goodness-of-fit for Luria-Delbruck distribution.

Figure 5: Statistical comparisons.

Conclusions

References

[1] Arnold, T. B. and Emerson, J. W. Nonparametric Goodness-of-Fit Tests for Discrete Null Distributions. *R Journal*, 3(2), 2011.

[2] M. Baum, S. Archontoulis, and M. Licht. Planting date, hybrid maturity, and weather effects on maize yield and crop stage. *Agronomy Journal*, 111(1):303–313, 2019.

[3] Brockman J.M. and Salaita, K. Mechanical proofreading: A general mechanism to enhance the fidelity of information transfer between cells. *Frontiers in Physics*, 7, feb 2019.

[4] Chiang, C.L. *Introduction to Stochastic Processes in Biostatistics*. John Wiley & Sons, Inc., 1968.

[5] F. Chmielewski, A. Müller, and E. Bruns. Climate changes and trends in phenology of fruit trees and field crops in germany, 1961–2000. *Agricultural and Forest Meteorology*, 121(1):69–78, 2004.

[6] C. Duden, C. Nacke, and F. Offermann. German yield and area data for 11 crops from 1979 to 2021 at a harmonized spatial resolution of 397 districts. *Scientific Data*, 11(1), Jan. 2024.

[7] Gillet-Markowska, A., Louvel, G., and Fischer, G. *bz-rates*: A Web Tool to Estimate Mutation Rates from Fluctuation Analysis. *G3 Genes|Genomes|Genetics*, 5(11):2323–2327, nov 2015.

[8] Jensen, J.D., Payseur, B.A., Stephan W., Aquadro, C.F., Lynch, M., Charlesworth D. and Charlesworth, B. The importance of the Neutral Theory in 1968 and 50 years on: A response to Kern and Hahn 2018. *Evolution*, 73(1):111–114, nov 2018.

[9] Z. Liu. *Crop Phenology: Timing is everything*, page 257–294. Springer Nature Singapore, 2026.

[10] Loeb, L.A., Springgate, C.F., and Battula, N. Errors in DNA Replication as a Basis of Malignant Changes. *Cancer Research*, 34(9):2311–2321, 09 1974.

[11] Luria, S.E. and Delbrück, M. Mutations of Bacteria from Virus Sensitivity to Virus Resistance. *Genetics*, 28(6):491–511, Nov 1943.

[12] Maby, P., Bindea, G., Mlecnik, B. and Galon, J. License to kill: microsatellite instability and immune contexture. *OncoImmunology*, 10(1), jan 2021.

[13] S. Mason, T. Galusha, and Z. Kmail. Planting date influence on yield of drought-tolerant maize with different maturity classifications. *Agronomy Journal*, 110(1):293–299, 2018.

[14] T. Miedaner and P. Juroszek. Global warming and increasing maize cultivation demand comprehensive efforts in disease and insect resistance breeding in north-western europe. *Plant Pathology*, 70(5):1032–1046, 2021.

[15] J. Olesen, C. Børgesen, L. Elsgaard, T. Palosuo, R. Rötter, A. Skjelvåg, P. Peltonen-Sainio, T. Börjesson, M. Trnka, F. Ewert, S. Siebert, N. Brisson, J. Eitzinger, E. van Asselt, M. Oberforster, and H. van der Fels-Klerx. Changes in time of sowing, flowering and maturity of cereals in europe under climate change. *Food Additives & Contaminants: Part A*, 29(10):1527–1542, Aug. 2012.

[16] R. Patil, M. Lægdsman, J. Olesen, and J. Porter. Growth and yield response of winter wheat to soil warming and rainfall patterns. *The Journal of Agricultural Science*, 148(5):553–566, 2010.

[17] P. Peltonen-Sainio, A. Kangas, Y. Salo, and L. Jauhiainen. Grain number dominates grain weight in temperate cereal yield determination: Evidence based on 30 years of multi-location trials. *Field Crops Research*, 100(2-3):179–188, Feb. 2007.

[18] Rohatgi, V.K. and Saleh, A.K.M.E. *An Introduction to Probability and Statistics*. Wiley Series in Probability and Statistics. John Wiley & Sons, Inc, 2015.

[19] Sherwood, L. and Ward, C. *Human physiology: from cells to systems*. Cengage learning, 2018.

[20] Sniegowski, P.D., Gerrish, P.J., Johnson T., and Shaver, A. The evolution of mutation rates: separating causes from consequences. *BioEssays*, 22(12):1057–1066, nov 2000.